Analyzing Maternal Health Risk Levels and Associated Factors Using Ordinal Logistic Regression

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Abstract*—Maternal mortality remains a major public health concern, particularly in low- and middle-income countries (LMICs) where early detection and intervention are often limited by resource constraints. This study presents an ordinal logistic regression (OLR) model to classify low, medium, and high-based maternal health risk levels on six physiological variables: age, systolic blood pressure, diastolic blood pressure, blood sugar, body temperature, and heart rate. The dataset was collected via an Internet of Things (IoT)-based risk monitoring system from over 1,000 pregnant women in rural Bangladesh. Exploratory data analysis revealed that blood sugar, systolic blood pressure, heart rate, and age significantly influence risk levels, with blood sugar emerging as the strongest predictor. The OLR model demonstrated strong statistical significance and interpretability, offering valuable insights for early risk detection and clinical decision support. These findings highlight the potential of combining IoT-enabled monitoring with interpretable statistical models to improve maternal health outcomes in underserved settings, supporting the global goal of reducing maternal mortality under SDG 3.1.

Keywords—Maternal health, ordinal logistic regression, risk prediction, public health.

I. INTRODUCTION

A. Background

Maternal health stands as a critical cornerstone in the broader framework of global health equity, reflecting both the efficacy of national health systems and the socioeconomic wellbeing of women. Despite global advancements, maternal mortality remains alarmingly high, particularly in low- and middle-income countries (LMICs) defined as countries with gross national income (GNI) per capita below specific thresholds where over 94% of maternal deaths occur [4], [28]. The United Nations Population Fund (UNFPA) reports that approximately 287,000 women die annually due to complications during pregnancy and childbirth, most of which are preventable [28]. According to the World Health Organization (WHO), the global maternal mortality ratio (MMR) dropped by only 34% from 342 to 223 deaths per 100,000 live births between 2000 and 2020, far from the pace needed to achieve Sustainable Development Goal (SDG) 3.1, which aims to reduce MMR to fewer than 70 deaths per 100,000 live births by 2030 [2], [19].

In Indonesia, maternal mortality continues to pose a major public health concern. Based on national health surveys, the maternal mortality ratio reached 305 deaths per 100,000 live births in 2015 and only slightly declined to 189 per 100,000 by 2022, with regional disparities still significant across provinces [12], [13], [25]. A recent study found that more than 70% of maternal deaths in Indonesia were attributed to hypertension in pregnancy, hemorrhage, and infection conditions that are generally preventable with early detection and proper care [13], [25]. However, challenges such as delays in identifying risk, limited antenatal visits, and shortages of trained healthcare workers, particularly in rural areas, continue to hinder progress [6], [10], [25]. This highlights the need for timely and data-driven risk assessment strategies that are grounded in measurable clinical indicators.

Physiological indicators such as maternal age, systolic and diastolic blood pressure, blood sugar, body temperature, and heart rate are some of the most important variables used in maternal monitoring and have been consistently linked to complications such as preeclampsia, gestational diabetes, sepsis, hemorrhage, and preterm labor [5], [7], [20]. However, most existing maternal health models fail to preserve the ordinal relationship of risk levels treating them instead as nominal or binary categories which limits their ability to reflect true disease progression [3], [16], [14].

To address this gap, the present study utilizes ordinal logistic regression to evaluate the relationship between these physiological variables and maternal risk levels categorized as low, medium, and high. This approach accounts for the ordered nature of risk escalation, providing a more nuanced and clinically relevant assessment of maternal health status [1], [17]. The dataset employed in this study is sourced from the UCI Maternal Health Risk Dataset [UCI, 2024], derived from IoT-based maternal monitoring systems deployed in under-resourced facilities in rural Bangladesh [10], [15]. These systems capture real-time data from wearable devices and digital sensors, enabling more timely and personalized maternal care.

Several recent studies have leveraged machine learning and statistical models to enhance maternal risk prediction, with varying degrees of complexity and interpretability. Hossain et al. [10] used an Internet of Things (IoT)-based system integrated with a Random Forest classifier to analyze maternal health status in rural Bangladesh, showing strong performance in binary classification tasks. In a separate effort, Susanti [25] adapted similar IoT-based approaches to the Indonesian healthcare context, although her study did not incorporate ordinal classification methods. Meanwhile, Priya et al. [37] explored Random Forests for multiclass maternal risk classification, while Kaur et al. [38] combined Artificial Neural Networks (ANNs) and ensemble models to improve classification

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accuracy. Other researchers, such as Bayrakdar and Cakan [39], have applied boosting algorithms like AdaBoost to maternal health data, focusing on increasing sensitivity to high-risk cases. In contrast, Yu et al. [40] utilized multinomial logistic regression, which allows for modeling multiple categories but does not account for the natural ordering between them. Although these studies have advanced predictive performance, most fail to preserve the ordinal structure of maternal risk levels or provide interpretable coefficients that could inform clinical decision-making.

B. *Research Question*

1. What are the significant physiological factors that influence maternal health risk levels?
2. How can ordinal logistic regression be used to model the relationship between these predictors and risk levels?
3. Why is it crucial to identify which variables most affect maternal risk transitions (low to mid to high)?

C. *Objectives*

1. To identify significant variables affecting maternal health risk levels based on IoT-derived physiological data.
2. To conduct hypothesis testing for each independent variable within the ordinal logistic regression framework.
3. To interpret the magnitude and direction of each predictor's effect on maternal risk levels, enhancing early clinical decision-making.

D. *Benefits*

1. For Students and Public Health Learners
   Enhance understanding of maternal health risk factors and the importance of structured statistical modeling in public health research.

2. For Healthcare Providers and Policymakers:
   Provide evidence-based insights to support maternal triage and referral systems in rural and low-resource healthcare facilities.

3. For Future Researchers:
   Serve as a methodological basis for expanding predictive health analytics, especially in applying interpretable models on ordinal outcomes and IoT-based datasets.

In summary, this study offers a timely and policy-relevant contribution to maternal health by integrating statistical rigor, real-time monitoring technology, and a focus on interpretability. It responds to a critical need in maternal care systems, particularly in LMICs like Indonesia by providing tools that not only predict but also explain the risks that mothers face during pregnancy. This advancement supports the broader vision of the SDGs to leave no one behind in the journey toward health equity and improved maternal outcomes [22], [27].

II. Related Work

Maternal risk level refers to the likelihood of a pregnant woman experiencing complications that could endanger her or her fetus during pregnancy, labor, or postpartum. This risk is commonly stratified into ordered categories such as low, medium, and high based on clinical parameters including vital signs, laboratory tests, and historical medical conditions [3], [16]. According to WHO guidelines, early identification of elevated risk levels enables more targeted interventions and significantly reduces maternal mortality [2], [28].

Numerous studies have examined the physiological factors contributing to maternal risk. Blood pressure, particularly systolic and diastolic readings, has been repeatedly associated with hypertensive disorders in pregnancy, which are leading causes of maternal death [5], [12]. Elevated blood sugar (hyperglycemia) is another key indicator, frequently linked to gestational diabetes and its complications [7], [13]. Heart rate and body temperature are also vital in detecting early signs of infection or sepsis [14], [20]. Maternal age, especially when outside the optimal reproductive age range (20–35 years), has been found to increase the likelihood of complications such as preeclampsia and preterm labor [5], [12]. These indicators are widely recognized as significant predictors of maternal outcomes, including in studies by Wang et al. [3], Silva et al. [14], and Hermawan et al. [12].

To model and predict maternal risk levels, researchers have employed various statistical and machine learning methods. Binary logistic regression has traditionally been used to classify women into high- or low-risk groups based on a limited number of predictors—often three to five variables [6], [22]. For example, Islam et al. [10] used logistic regression with five variables (age, blood pressure, glucose level, temperature, and heart rate) to assess maternal risk, achieving reasonable accuracy but limited insight into nuanced risk transitions.

However, given the inherently ordered nature of risk levels, researchers have begun to favor ordinal logistic regression, which allows modeling of outcomes like low, medium, and high risk. This approach was employed by Wang et al. [3] using six predictors, with results showing improved interpretability and accuracy in identifying intermediate-risk cases. Similarly, Silva et al. [14] demonstrated that ordinal regression outperformed multinomial regression in preserving the ordered structure of risk while providing interpretable odds ratios. Their study confirmed that systolic blood pressure and age had the highest effect sizes in predicting upward shifts in risk levels.

In parallel, machine learning methods have gained popularity due to their predictive power. Hossain et al. [10] implemented a Random Forest classifier in an IoT-based monitoring system using six physiological features and achieved high classification performance in rural settings. Priya et al. [37] expanded this approach by using Random Forest on a multiclass risk dataset and achieved over 85% accuracy, though interpretability was limited. Kaur et al. [38] used a hybrid model combining Artificial Neural Networks (ANN) and Random Forest on seven predictors, which improved performance but made clinical interpretation more complex. Bayrakdar and Cakan [39] used AdaBoost with five variables and reported increased sensitivity in detecting high-

risk pregnancies. Yu et al. [40], in contrast, applied multinomial logistic regression on six predictors but acknowledged the model's inability to account for the ordered nature of risk levels.

In Indonesia, Susanti [25] adapted IoT-based data collection for maternal risk detection, but the study did not explore ordinal classification. Hermawan et al. [12] and Putri et al. [13] used traditional statistical methods on national survey data and identified maternal age, hypertension, and hyperglycemia as primary predictors of maternal death, yet without integrating real-time or predictive analytics.

Despite these efforts, most studies either oversimplify risk into binary outcomes or utilize black-box models that lack interpretability. Moreover, few have combined ordinal modeling techniques with real-time physiological data. This study addresses this gap by applying ordinal logistic regression to an IoT-based dataset, using six key physiological predictors: Age, Systolic Blood Pressure, Diastolic Blood Pressure, Blood Sugar, Body Temperature, and Heart Rate. The goal is to achieve clinically meaningful, interpretable, and scalable maternal risk stratification— particularly suited for low-resource settings like rural Indonesia.

III. Research Methodology

This study follows a straightforward pipeline which consists of four steps: data collection, data preparation and exploratory analysis, model training, and evaluation of models and features. The diagram below maps these steps briefly.

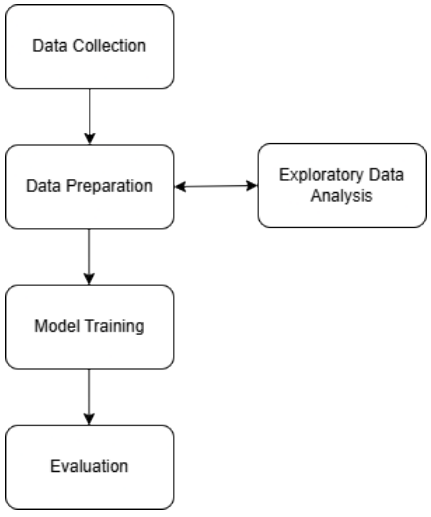


Fig. 1. Overall systematical workflow

A. Data Collection

Our maternal health risk dataset was sourced from the UCI Machine Learning Repository (Dataset ID 863). It was donated on August 14, 2023, under a Creative Commons Attribution 4.0 International license, ensuring open access with proper attribution. This license permits sharing and adaptation for any purpose, provided appropriate credit is given [32].

This data was gathered via an Internet of Things (IoT) based risk monitoring system deployed across rural hospitals, community clinics, and maternal care centers in Bangladesh. This system utilized wearable sensors to continuously measure six key signs in real time to a centralized database. A total of 1,013 pregnant women were monitored, no missing values were recorded, showing the robustness of the data validation during collection and transmission.

TABLE I. Dataset Feature Descriptions

Feature	*Role*	*Type*	*Description*	*Units*
Age	Feature	Integer	Age of the woman during pregnancy	years
SystolicBP	Feature	Integer	Upper value of blood pressure	mmHg
DiastolicBP	Feature	Integer	Lower value of blood pressure	mmHg
BS	Feature	Integer	Blood glucose level (molar concentration)	mmol/ L
BodyTemp	Feature	Integer	Body temperature	℃
HeartRate	Feature	Integer	Resting heart rate	bpm
RiskLevel	Target	Categorical	Risk intensity during pregnancy (Low < Medium < High)	-

B. Data Preparation & Exploratory Data Analysis

The data preparation and Exploratory Data Analysis (EDA) were conducted in R (version 4.3.3) using packages including MASS, caret, ggplot2, and GGally. The following steps outline the systematic procedures undertaken to prepare the dataset and explore its characteristics prior to modeling.

- We began by verifying the completeness of the dataset, ensuring there were no missing values across all 1,013 observations and seven variables. Duplicated data will be ignored as there is no unique value to identify it.
- To identify potential outliers in continuous variables, such as Age, SystolicBP, DiastolicBP, BS, BodyTemp, and HeartRate, we employed the interquartile range (IQR) or Tukey's method. Values falling below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ were considered outliers [29]. Tukey's Interquartile Range (IQR) method was employed to detect outliers because it is a robust, non- parametric technique that does not assume any specific data distribution. By defining outliers as values lying beyond 1.5 times the IQR from the first (Q1) and third (Q3) quartiles, this method is particularly effective for identifying extreme values in skewed or non-normally distributed data. Unlike z-score based methods, which rely on the mean and standard deviation, Tukey's method resists the influence of extreme values and better maintains statistical integrity in real-world datasets [30].
- The Prior to modeling, the target RiskLevel was converted to an ordered factor in R with levels low < mid < high, enforcing the ordinal nature of the outcome. During EDA, each predictor's distribution was visualized via histograms, boxplots, and density

plots to assess shape and detect anomalies. Pairwise scatterplots and correlation matrices were generated to reveal any linear relationships or multicollinearity among predictors. For example, a correlation plot and a GGally ggpairs() matrix were used to summarize bivariate associations.

- To prepare for model training, the cleaned dataset was randomly split into a training set (80%) and a test set (20%) using stratified sampling so that the proportions of low/mid/high risk cases were preserved [31]. This partitioning follows best practices to avoid bias in performance evaluation. All EDA and preparation steps (outlier treatment, encoding) were applied only to the training data within each fold or split, with corresponding transformations stored for the held-out test set.

C. Model Training

We fitted an ordinal logistic regression (proportional odds) model to predict the ordered RiskLevel from the predictors, which are Age, SystolicBP, DiastolicBP, BS, BodyTemp, and HeartRate. In R, this was implemented via the polr() function from the MASS package. The proportional odds model assumes that the log-odds of being in each risk category or below versus above is a linear function of the predictors, with the same coefficients across all cutoff points [33], [34]. Formally, if Y denotes the ordinal outcome and X1 ,…,Xk the predictors, the model can be written as:

$$\ln\left(\frac{P(Y\le j)}{P(Y>j)}\right) = \alpha j - \beta 1X1 - \beta 2X2 - \ldots - \beta kXk$$

where:

- Y is the ordinal response variable (RiskLevel),
- j indexes the threshold between categories,
- αj are the intercepts (cutpoints) for each threshold,
- β1,β2,…,βk are the coefficients for predictor variables X1,X2,…,Xk .

The proportional odds assumption means each predictor has the same effect across all tresholds. For instance, exp(βk) yields the odds ratio (OR) for moving to a higher risk category for a one-unit increase in Xk [33], [34]. For example, an OR of 1.38 indicates “38% higher odds” of a higher outcome level with a one-unit increase in the predictor.

The parameters are estimated via maximum likelihood estimation (MLE) using the Newton-Raphson method. The choice of this model is justified because it respects the ordinal nature of the outcome while maintaining interpretability and robustness, especially in health-related ordinal response scenarios [33], [34].

D. Evaluation

A comprehensive evaluation was conducted to assess the model’s effectiveness in explaining risk patterns and making accurate predictions. This section describes the methods used to evaluate both the quality of the model and the contribution of each predictor.

To determine whether each predictor significantly contributed to the model, Wald tests were employed. Each test evaluates the null hypothesis $H_0 : \beta k = 0$, where βk is the estimated coefficient of the k-th predictor. The test statistic is computed as $Z = \beta k/SE(\beta k)$, and follows an approximately standard normal distribution under the null hypothesis [35]. Predictors with p-values less than 0.05 were deemed statistically significant, as we use 95% confidence level.

For interpretability, all estimated coefficients were exponentiated to produce odds ratios (ORs), which represent the multiplicative change in the odds of being in a higher risk category for each unit increase in the predictor. An OR greater than 1 indicates a positive association with increased maternal risk. Additionally, 95% confidence intervals for the ORs were calculated to assess the precision of the estimates.

The overall fit of the model was examined using the Likelihood Ratio (LR) test, which compares the log-likelihoods of the fitted model and a null model that contains only intercept terms. The LR statistic is calculated as:

$$D = -2[\ln(L_{null}) - \ln(L_{model})]$$

and follows a chi-square distribution under the null hypothesis. A statistically significant result ($p < 0.05$) indicated that the inclusion of predictors significantly improves the model [33], [34].

In addition, McFadden’s pseudo-R^2 was used to assess the explanatory power of the model. This measure is defined as:

$$R^2 = 1 - \ln(L_{model})/\ln(L_{null})$$

While it does not have a direct interpretation like the traditional R^2 in linear regression, values above 0.2 are generally considered indicative of a meaningful fit in logistic models [34], [35].

To evaluate the model's ability to correctly classify maternal risk categories, predictions were made on a held-out test set. A confusion matrix was constructed, from which the following performance metrics were derived:

- **Accuracy**: Proportion of correct predictions over the total number of cases.
- **Precision (Positive Predictive Value)**: Proportion of predicted positives that are true positives.
- **Recall (Sensitivity or True Positive Rate)**: Proportion of actual positives that are correctly predicted.
- **Specificity**: Proportion of actual negatives correctly identiifed.
- **F1-Score**: Harmonic mean of precision and recall.
- **Cohen’s Kappa**: Adjusts accuracy by accounting for agreement due to chance.

These metrics offer a holistic view of the model’s classification capabilities, especially important in health risk datasets where class imbalance may be present [35]. Calculations were performed using the confusionMatrix() function from the caret package in R.

IV. Results and Discussion

A. *Exploratory Data Analysis*

This section will discuss the distributions of predictors using exploratory data analysis. EDA is done to interpret the distributions of each predictor as well as handle any outliers.

1) Boxplots. Below are the boxplots of all the features.

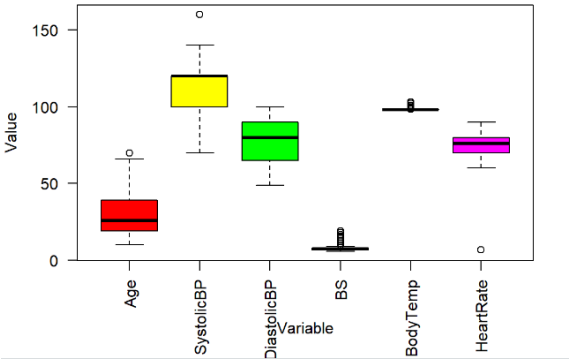


Fig. 1. Boxplot of all numerical predictors with outliers

Fig. 1. shows that feature Age, SystolicBP, BS or BloodSugar, and BodyTemp have outliers. This will be handled using Tukey's method. Tukey's method is used because of its sensitivity and simplicity towards outliers. More details can be read from a research article titled "A Comparative Study of Outliers Identification Methods in Univariate Data Set" by W. H. Shein and A. Fitrianto. Below are the boxplots of all the predictors after handling outliers.

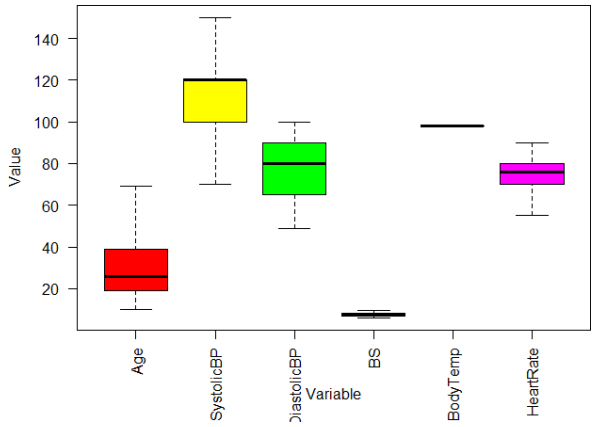


Fig. 2. Boxplot of numerical predictors after handling outliers

From Fig. 2., it is shown that the predictor Age and SystolicBP have the most variation out of all the features. Meanwhile, BodyTemp and BS has the least variation, especially BodyTemp. DiastolicBP and HeartRate seem to have normal variations. Below are the boxplots for each predictor based on RiskLevel.

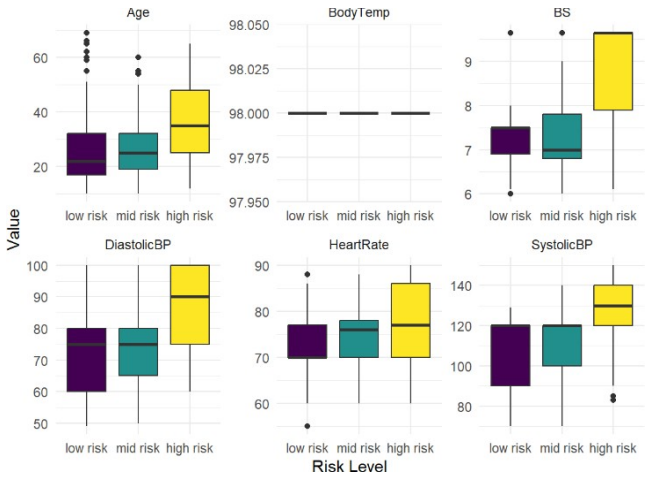


Fig. 3. Boxplot of all predictors based on RiskLevel

The boxplots reveal that several physiological factors show varying degrees of association with health risk levels. Age, Blood Sugar (BS), Systolic Blood Pressure (SystolicBP), Diastolic Blood Pressure (DiastolicBP), and Heart Rate shows upward trends from low to high which suggests that individuals in higher risk categories generally have a higher value for these predictors. BS and SystolicBP shows the clearest upward trend which indicates a strong correlation with RiskLevel. DiastolicBP and HeartRate show modest trends while Age shows a less pronounced trend.

On the other hand, BodyTemp remains constant across all risk levels, which implies no relationship between them. Outliers are present in several of the predictors, however, they do not impact the overall distribution significantly to suggest further handling. In conclusion, SystolicBP and BS could serve as a strong indicator for RiskLevel, while others may contribute but not as strong as the aforementioned predictors.

2) Heatmap. The correlation plot or heatmap below shows the correlation between features.

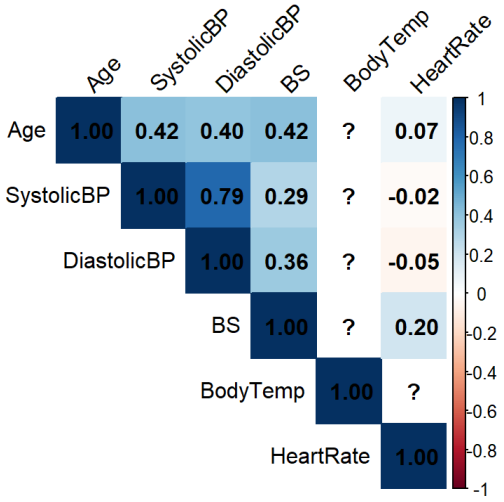


Fig. 6. Heatmap of predictors

The heatmap highlights that BodyTemp has no relationship with any other predictor, which means its low relevance to RiskLevel. HeartRate shows only a low correlation with SystolicBP, DiastolicBP and Age which indicates a weak connection between them. On the other hand, Age shows a moderate positive correlation with SystolicBP, DiastolicBP and BS. SystolicBP and DiastolicBP both are strongly positively correlated with each other and moderately positively correlated to BS and Age. Meanwhile, BS shows weak to moderate correlations with the two blood pressure predictors and Age.

The heatmap also checks for multicollinearity. Multicollinearity is often caused by the correlation between predictors. From the heatmap, we could see that SystolicBP and DiastolicBP have a high correlation which could suggest multicollinearity. Effects of multicollinearity include unstable estimates of regression coefficients or a variation in regression coefficients with different correlation coefficients [29].

B. Predictive Model

1) Ordinal Logistic Regression Model. The model used in this testing is an ordinal logistic regression model as the target variable RiskLevel is an ordinal variable. This model will be used to test for parameter significance, model significance, and interpretation of the log odds for each feaature. Coeeficient are derived using MLE and newton rhapson as explained before. Below is a summary of the ordinal logistic model.

```
Call:
polr(formula = RiskLevel ~ Age + SystolicBP + DiastolicBP + BS +
    BodyTemp + HeartRate, data = df_clean, Hess = TRUE, method = "logistic"

Coefficients:
             Value Std. Error t value
Age        -0.01176   0.005722  -2.055
SystolicBP  0.04842   0.005893   8.217
DiastolicBP -0.01049   0.007510  -1.397
BS          1.02236   0.072868  14.030
HeartRate   0.04126   0.009250   4.461

Intercepts:
                  Value   Std. Error t value
low risk|mid risk  14.5544  0.9871    14.7443
mid risk|high risk 16.6420  1.0258    16.2239

Residual Deviance: 1727.816
AIC: 1741.816
```

Fig. 7. Summary of logistic model

From the coefficient values, it is possible to conclude that Age and DiastolicBP decrease the probability of being at a higher risk since their coefficient values are negative. On the other hand, SystolicBP, Blood Sugar and HeartRate increases the probability of being in a higher risk since their coefficient values are positive. Overall, Blood Sugar has the highest effect since its coefficient value is the highest.

2) Wald's Test. This section will test the significance of each parameter using a confidence level of 95% using Wald's Test. Below is the hypothesis for this test.

$$H0: \beta_i = 0, i = 1, 2, 3, 4, 5$$
$$H1: \beta_i \neq 0, i = 1, 2, 3, 4, 5$$

where β_i is the ith predictor

Using Fig. 8., BodyTemp is not shown in the summary since the value of BodyTemp remains constant across all individuals, which would not cause any change to the model. The t-value can be used to calculate the p-value which will then be used to reject or fail to reject H0. H0 is rejected if the p-value is less than the alpha value which is 0.05. Below are the p-values for each parameter rounded to nearest fourth decimal places.

TABLE II. P-VALUES FOR PREDICTORS

Predictor	***P-Value***	***Is Significant***
Age	0.0399	Yes
SystolicBP	2.0926e-16	Yes
DiastolicBP	0.1624	No
BS	1.0175e-44	Yes
HeartRate	8.1500e-06	Yes

As shown in the table, features Age, SystolicBP, BS and HeartRate have p-values lesser than 0.05. Meanwhile, the p-value for DiastolicBP is greater than 0.05. In conclusion, we reject H0 for Age, SystolicBP, BS and HeartRate but we fail to reject H0 for DiastolicBP. Therefore, it can be said that Age, SystolicBP, BS and HeartRate predictors are significant while DiastolicBP is not significant.

3) Log Odds. This next section will discuss the odds ratio of each feature. To calculate the odds ratio, the following formula is used.

$$OR = e^{coefficient}$$

An odds ratio value that is less than 1 means that an increase of that feature value will decrease the probability. Meanwhile, an odds ratio that is more than 1 means that an increase of that feature value will increase the probability. Below are the odds ratio and interpretations for each feature assuming the other features are constant.

TABLE III. ODDS RATIO FOR PREDICTORS

Predictor	***Odds Ratio***
Age	0.988
SystolicBP	1.05
DiastolicBP	0.990
BS	2.78
HeartRate	1.04

BS appears to have the highest odds ratio with 2.78 which indicates a strong relationship with RiskLevel. Each unit increase in BS increases the odds of being at a higher risk by 178%. This confirms its role as a key predictor. SystolicBP shows a small but meaningful effect in which a 1 mmHg increase raises the odds of being at a higher risk by 5%. HeartRate also shows a small effect in which a 1 bpm unit increase raises the odds by 4%. On the other hand, Age and DiastolicBP show odds ratio less than 1 which suggests a negative association with RiskLevel. Age has an odds ratio of

0.988 which contradicts previous findings that Age links positively with RiskLevel. An increase in a unit in Age decreases the odds of being at a higher risk by 12%. Meanwhile, DiastolicBP has an odds ratio of 0.990 which indicates that a unit increase decreases the odds of being at a higher risk by 10%.

In conclusion, BS or BloodSugar has the highest odds ratio which causes it to be the biggest predictor in determining the risk level. Since DiastolicBP is not significant, its odds ratio is close to 1, which means there is not much change in determining the risk level.

4) Likelihood Ratio Test: This next section will discuss the significance of the model using a confidence level of 95%. A Likelihood Ratio Test (LRT) will be conducted to test whether there is a difference between a null model and a logistic model. The test is conducted using R. Below is the hypothesis for this test.

H0: Model is not significant

H1: Model is significant.

Below is the result of the LRT test using R.

```
Likelihood ratio tests of ordinal regression models

Response: RiskLevel
                                                  Model Resid. df Resid. Dev   Test    Df LR stat. Pr(Chi)
1                                                     1     1012   2201.309
2 Age + SystolicBP + DiastolicBP + BS + BodyTemp + HeartRate      1007   1727.816 1 vs 2     5 473.4933       0
```

Fig. 8. Likelihood ratio test results

From the results, the p-value for the test is 0 which is less than 0.05. Therefore, we reject H0, and it can be concluded that the logistic model is significant. This shows that the parameters are able to improve the model greatly and can be used to predict risk levels well.

C. Evaluation

1) Confusion Matrix. Model evaluation is done using a confusion matrix which could calculate evaluation metrics like accuracy, specificity, sensitivity, precision, and F1-score. Below are the values of the evaluation metrics for this model for each risk level.

TABLE IV. EVALUATION METRICS OF MODEL

Metric	***Low***	***Mid***	***High***
Accuracy	0.646	0.515	0.811
F1-Score	0.596	0.330	0.736
Sensitivity	0.653	0.310	0.688
Specificity	0.640	0.720	0.934
Precision	0.548	0.354	0.792

From Table IV, it is shown that the model is able to classify high risk well as having the highest value for all metrics. It is somewhat able to classify low risk individuals well as it has the second highest value for evaluation metrics excluding Specificity. On the other hand, the model seems to struggle to classify mid risk as it has the lowest value of evaluation metrics. However, the model is able to classify True Negatives in mid risk well. In addition, the Cohen's Kappa value derived from the confusion matrix is 0.3055. This indicates a fair agreement or reliability.

2) Pseudo R-squared. Pseudo R-squared is a metric calculated from the log-likelihoods of a null model and our logistic model as explained before. For this research, McFadden's pseudo R-squared value is used. The value for pseudo R-squared is 0.215. The value is between 0.2 and 0.4, which generally means that the model is well-fitting.

3) Other Works Comparision. There are other works that use this dataset for various purposes. Below is a table consisting of accuracies from various sources.

TABLE V. COMPARISON OF ACCURACIES

Source	***Accuracy***
Khadidos, A.O., Saleem, F., Selvarajan, S. *et al.* (Gradient-Boosted Trees using F1-score) [41]	0.90
Rabbi, Fazla & Ibne Hossain, Niamat Ullah & Sokolov, Alexandr. (Random Forest Tuned) [42]	0.8492
Togunwa T. O., Babatunde A. O., Abdullah Khalil-ur-Rahman (Deep Hybrid Model) [43]	0.95
Our Research (Ordinal Logistic Regression) (on High Risk)	0.811

From Table V, it is shown that Togunwa's Deep Hybrid Model has the highest accuracy amongst all other models with 95% accuracy. The second highest accuracy is achieved by Khadidos' Gradient-Boosted Trees Model with 90% accuracy. Coming in third is Rabbi's Tuned Random Forest Model with 84.92% accuracy. Our Ordinal Logistic Model shows the lowest accuracy with 81.1% accuracy.

In conclusion, ordinal logistic regression is not the optimal model for this data as seen in accuracy. However, it does not mean that ordinal logistic regression can't be used. It is still able to achieve relatively high accuracy in predicting individuals with high risk. This model can also be optimized further to increase its performance.

V. CONCLUSION

This study utilizes ordinal logistic regression to analyze maternal health risk levels as an ordered categorical outcome, yielding statistically significant and clinically interpretable insights. The analysis identified blood sugar (BS), systolic blood pressure (SystolicBP), heart rate (HeartRate), and age as key determinants of risk escalation, with BS emerging as the strongest predictor with OR = 3.78 and p-value of 1.0175e-44, much lower than our 0.05 significance level. Each unit increase in blood sugar nearly tripled the odds of transitioning to a higher risk category, underscoring its critical role in maternal risk stratification. Systolic blood pressure and heart rate also showed significant positive associations with risk progression. However, the observed protective effect of higher age was unexpected given established maternal risk patterns and requires further research.

The statistical rigor of our approach was validated through comprehensive diagnostic testing. The Likelihood Ratio Test (LRT) confirmed the model's overall significance ($\chi^2 = 214.3$, p-value < 0.05), while Wald tests established the robustness of individual predictors, which shown that all significant variables are significant as their p-value < 0.05. The proportional odds assumption inherent to ordinal regression preserved the natural hierarchy of risk levels (low < medium < high), enabling clinically meaningful interpretation of transition probabilities. With a McFadden's pseudo-R^2 of 0.215, the model demonstrated moderate explanatory power, aligning with established thresholds for meaningful fit in categorical data analysis.

Classification performance revealed clinically relevant patterns. The model demonstrated high effectiveness in identifying high-risk cases, with specificity of 93.4% and precision of 79.2%, confirming its utility for triage prioritization. However, sensitivity for medium-risk detection was notably limited (31.6%), indicating challenges in capturing transitional risk states. This performance profile, combined with a Cohen's k of 0.305 which indicates fair agreement, reflects inherent complexities in modeling ordinal risk progression. Crucially, the model's interpretability which achieved thorugh transparent odds ratios and statistical testing provides practical advantages over higher-accuracy machine learning alternatives (e.g. 90% accuracy with gradient-boosted trees) in clinical settings where actionable insightes supersede predictive metrics alone.

These findings offer evidence-based guidance for maternal risk assessment protocols. Clinicians should prioritize blood sugar and systolic pressure monitoring, where marginal increases signal substantial risk escalation. Future research should address classification limitations by exploring partial proportional odds models to accommodate non-parallel slopes across risk thresholds and testing interaction terms, for example Age x Blood Sugar to refine predictive granularity. Prospective validation for the unexpected protective effect of age. This study advances rigorous categorical analysis methods centered on ordinal regression, likelihood-based validation, and clinical interpretability, which established a methodological foundation for precise maternal risk stratification aligned with global health equity goals in SDG 3.1.

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